The Photonic Holonet

A Single Self-Entangled Photon as Universal Computer, Universal Network, and Clock on the $W(3,3)$ Substrate

Wil Dahn

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Abstract

We present a complete, machine-verified architecture in which one self-entangled photon, traversing one finite geometry, is simultaneously the hardware, the software, and the network of a universal quantum computer. The geometry is the symplectic generalized quadrangle $W(3,3)$: forty points, forty lines, automorphism group $\mathrm{Sp}(4,\mathbb{F}_3)$ of order 51840. The photon carries $W(3,3)$ twice over — once as the forty Witting rays of its path–polarization space $\mathbb{C}^4$ (states), once as the forty Pauli displacement classes of its past–future time-bin space $\mathbb{C}^9$ (operators) — and the two carriers realize the same incidence geometry, making the machine's states and gates one set of objects. Every architectural layer is a theorem with an executable witness: the preparation circuit (two Clifford gates of tabletop optics), the routing fabric (an atlas of 540 hypercube charts glued along the 1620 apartments of the Tits building), the protected memory (the Steinberg module of dimension 81), the mirror middleware (a 2160-slot D_{12} transport bus whose full Clifford lift factors as $24 \cdot 45 \cdot 48 = 51840$), the fuel (the matter shell is exactly the magic sector, with exact Kochen–Specker budget 36/40 and contextual fraction 1/10), the clock (an internal $\mathbb{Z}_{12}$ gauge clock, two external references $\mathbb{Z}_7, \mathbb{Z}_{13}$, and an irrational Boerdijk–Coxeter drive realizing a discrete time quasicrystal), and universality (the photon's own tritter, phase plate, and modulator generate the complete Clifford group; the matter shell supplies the magic). We also give the recursive engineering law: a level-n holonet has 40^n photonic leaves, $(40^n - 1)/39$ $W(3,3)$ instances, and reversible routing diameter bounded by $8n = 8\log_{40} N$, while durable commits use a separate Császár/tomotope clock $T(n) = 4(7^n - 1)$. The newest runtime layer turns the tomotope cover pathology into an engineering feature: durable storage is cover-indexed by a $\mathbb{Z}_k^3$ fiber over the stable 48-block packet ABI, sentinel-aware compilation can trade commit-time slack for lower $g = 15$ fault energy, the first exact guard band is the arithmetically forced desynchronization at $n = 5$ with remainder $24 = f$, and the Grünbaum–Coxeter Wythoff operations of the tomotope are exactly the packet-growth ladder $4, 12, 24, 48, 96$. No fitted parameters appear anywhere: every constant is substrate arithmetic at $q = 3$.

Contents

1 Overture: the machine is the geometry is the carrier

Classical architecture separates three things: the processor that acts, the program that directs, and the network that connects. The thesis of this paper is that at the level of a single photon on a finite symplectic geometry the three are one object, and that this is not a metaphor but a chain of theorems.

Principle 1.1 (Identity of the three layers)**.** *Transporting the photon through the mesh* is *applying a gate* is *routing a packet. One physical process; three descriptions.*

The identification rests on five exact pillars, each proved by explicit computation in the project repository (every theorem below carries its witness script):

1. **Operator–operand duality** (§2.1): the same $W(3,3)$ is drawn by the photon's *states* (orthogonality of the 40 Witting rays in $\mathbb{C}^4$) and by its *operators* (commutation of the 40 two-qutrit Pauli classes on $\mathbb{C}^9$).

2. **Routing–gate identity** (§4): the mesh is an atlas of 540 three-dimensional hypercube charts whose native XOR routing moves are exactly the register operations, glued along the 1620 apartments of the Tits building of $\mathrm{Sp}(4, \mathbb{F}_3)$.

3. **Mirror middleware** (§5): the cyclic selector clock and the mirror transport bus are different 2160 worlds. The photon is clocked by the C_{12} side but routed by the D_{12} side, whose full Clifford lift is the exact product $24 \cdot 45 \cdot 48$.

4. **Universality from optics** (§6.5): the photon's own beam-splitting elements generate the complete Clifford group (symplectic closure = 51840, exact), and its matter shell is precisely its magic supply.

5. **Fractal holonet scaling** (§7): recursively replacing every site by a fresh $W(3, 3)$ core gives a universal computer/network with exact 40^n leaves and logarithmic reversible routing.

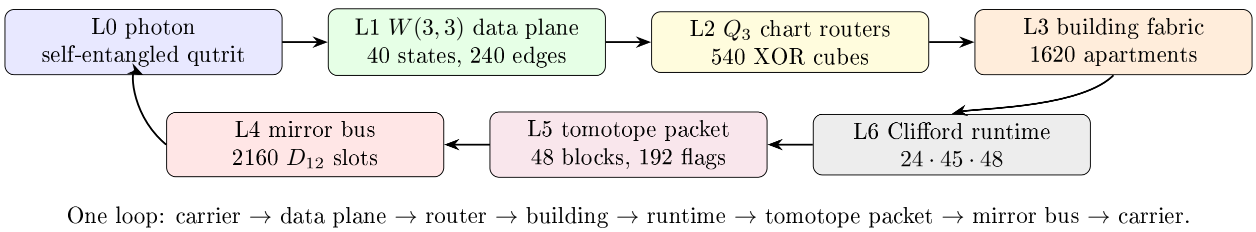


Figure 1: The photonic holonet as an engineering stack. The same photon is data, instruction, and packet; the same finite geometry supplies the network, middleware, and runtime.

2 The substrate

Definition 2.1 ($W(3, 3)$)**.** *Let $V = \mathbb{F}_3^4$ with the alternating form $\langle u, v\rangle = u_1v_3 - u_3v_1 + u_2v_4 - u_4v_2$. The points of $W(3, 3)$ are the 40 projective points of V; the lines are the 40 totally isotropic projective lines. Each point lies on 4 lines and each line carries 4 points; collinearity defines the strongly regular graph* $\mathrm{SRG}(40, 12, 2, 4)$.

The full automorphism group is $\mathrm{Sp}(4, \mathbb{F}_3)$, of order $51840 = |W(E_6)|$, acting with permutation rank 3. The substrate primitives are

$$q = 3, \quad \lambda = 2, \quad \mu = 4, \quad k = 12, \quad f = 24, \quad g = 15, \quad \Phi_6 = 7, \quad \Phi_4 = 10, \quad \Phi_3 = 13,$$

and every numerical claim in this paper reduces to arithmetic in these.

Formal grounding (the master formalism). In the master manuscript [1] the whole structure is forced by one Diophantine seed: the *master equation* $q! = 2q$ has the unique positive-integer solution $q = 3$, and the SRG quadratic $x^2 - q!\,x + 2^q = (x - 2)(x - 4)$ then delivers $\lambda = 2$, $\mu = 4$ and the full parameter closure ($k = 2^q + q + 1$, $v = 40$, multiplicities $f = 24$, $g = 15$, cyclotomic ladder $\Phi_3 = 13$, $\Phi_6 = 7$, $\Phi_{12} = q^4 - q^2 + 1 = 73$, $\Theta = q^2 + 1 = 10$). Two of those closure constants reappear in this paper as *spectral field discriminants*: the chart-web's irrational eigenvalue pairs $1 \pm \sqrt{10}$ and $(-1 \pm \sqrt{73})/2$ (§4) live precisely in $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ — the machine's transport spectrum speaks the same cyclotomic dialect as the physics tier ladder.

2.1 The two carriers and the operator–operand duality

Theorem 2.2 (Two realizations, one geometry)**.** (i) *The* 40 *Witting rays in* $\mathbb{C}^4$ *(the photon's path*$\otimes$*polarization space) have orthogonality graph isomorphic to the collinearity graph of* $W(3,3)$*.* (ii) *The* 40 *projective classes of two-qutrit Pauli displacements* $D(v) = X^aZ^b \otimes X^cZ^d$ *on* $\mathbb{C}^9$ *(the photon's past*$\otimes$*future time-bin space) have commutation graph equal to the same* $W(3,3)$*, via* $D(u)D(v) = \omega^{\langle u,v\rangle}D(v)D(u)$*.* Witnesses: `bt817_self_entangled_photon_atlas.py`, `bt821_operator_operand_duality.py`.

Thus a point of the substrate is simultaneously a pure state the photon can occupy and a unitary the photon can enact; a line is simultaneously a measurement context (orthonormal tetrad) and a stabilizer group (maximal commuting Pauli set with its 9-state joint eigenbasis). The $360 = 40 \times 9 = f \cdot g$ two-qutrit stabilizer states are the joint eigenbases of the forty lines.

2.2 The five vacua and the Schläfli geography

Every maximal subgroup class of $\mathrm{PSp}(4,3)$ decomposes the substrate in its own canonical way; the maximal indices are the classical inventory of the 27 lines on a cubic surface.

index	subgroup	points	lines	vacuum	cubic surface
27	$2^4{:}A_5$	$[40]$	$[40]$	homogeneous (icosahedral)	27 lines
36	S_6	$[40]$	$[10,30]$	spread fibration	36 double-sixes
40	parabolic	$[1,12,27]$	$[4,36]$	holonet point vacuum	trihedra I
40	parabolic	$[4,36]$	$[1,12,27]$	dual line vacuum	trihedra II
45	$(2T \times 2T){:}2$	$[8,32]$	$[16,24]$	binary polar	45 tritangents

The physical decomposition $40 = 1+12+27$ (self + gauge shell + matter shell) is the *choice of the point-parabolic vacuum* among five inequivalent readings; the icosahedral vacuum sees a featureless substrate. The platonic ladder threads the table: the binary tetrahedral group $2T$ (the 24-cell) lives in every polar-pair stabilizer, the real octahedral group O_h ($= S_4 \times \mathbb{Z}_2$, order 48) is the chart symmetry of §4, and the binary icosahedral group $2I = \mathrm{SL}(2,5)$ (the 600-cell) is the core of every spread stabilizer, with point orbits $[20,20]$ and line orbits $[10,30]$ echoing the Boerdijk–Coxeter decomposition $600 = 20 \times 30$. *Witnesses:* `bt808-bt813`*.*

3 The carrier: how to self-entangle a photon

Entanglement is a relation between tensor factors; nothing requires the factors to be distinct particles. Self-entanglement is entanglement between one photon's own registers.

3.1 Stage A: spatial (one element)

A diagonal photon meets a polarizing beam splitter: $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \;\mapsto\; \frac{1}{\sqrt{2}}(|H,a\rangle + |V,b\rangle)$. Path and polarization are now maximally entangled *within* the photon; this two-qubit $\mathbb{C}^4$ is the Witting carrier of Theorem 2.2(i).

3.2 Stage B: temporal (two Clifford gates)

A tritter (symmetric 3-port coupler) *is* the qutrit Fourier gate F_3; a three-bin delay ladder $(0,\tau,2\tau)$ defines past and future registers; one electro-optic modulator applies $\mathrm{CX}_{p\to f}$ conditioned on bin index. The temporal Bell qutrit

$$|\Omega\rangle \;=\; \mathrm{CX}_{p\to f}\,(F_3\otimes I)|0\rangle_p|0\rangle_f \;=\; \frac{1}{\sqrt{3}}\sum_{j=0}^{2}|j\rangle_p|j\rangle_f$$

is verified exactly; the photon is entangled with its own future. Inserting any U in the future arm, the recombination visibility is the *trace–Choi witness*

$$V(U) \;=\; \frac{|\mathrm{Tr}\,U|}{3}, \qquad V(F_3)=\tfrac{1}{3}, \quad V(X)=V(Z)=0,$$

so the photon implements the Choi–Jamiołkowski isomorphism on itself: it measures channels against its own past. *Witness:* `bt820_self_entanglement_protocol.py`.

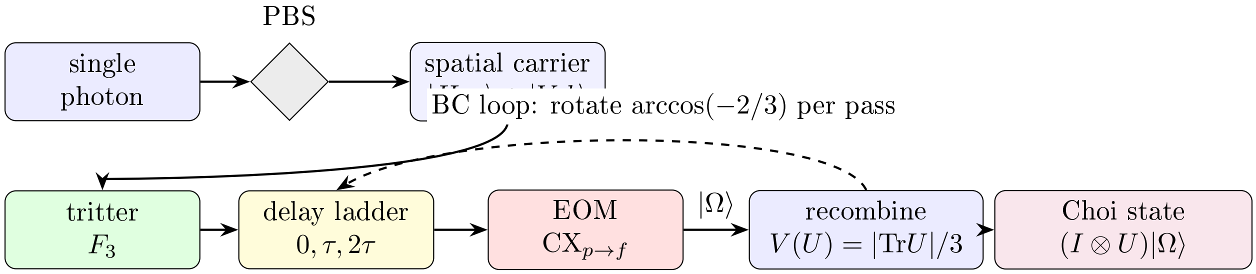


Figure 2: Preparation schematic. One photon, four registers. The PBS entangles path with polarization (Stage A); the tritter–delay–EOM chain prepares the temporal Bell qutrit $|\Omega\rangle$ (Stage B); the dashed loop is the irrational Boerdijk–Coxeter drive of §11.

4 The network: atlas, building, and routing

Theorem 4.1 (The hypercube atlas)**.** *$W(3,3)$ contains exactly 540 skew line pairs. For each pair the non-collinearity graph on its 8 points is the cube graph Q_3, with full symmetry group O_h of order 48 (these are the only order-48 stabilizers; the binary $2O$ does not occur). Each chart admits an exact $\mathbb{F}_2^3$ Gray-code addressing in which chart edges are unit XOR moves and the antipode of a vertex is its unique collinear partner. Every $W(3,3)$ nonedge is a chart edge in exactly $12=k$ charts; every edge is an antipode pair in exactly $9=q^2$.* Witnesses: `bt773`, `bt777`, `bt778`.

4.1 Hypercube networking, not hypercube decoration

The cube is not only a picture for one chart. It is the local interconnect primitive. A chart gives eight simultaneously valid addresses

$$000,\ 001,\ 010,\ 011,\ 100,\ 101,\ 110,\ 111 \in \mathbb{F}_2^3,$$

and the elementary network instruction is the dimension-order hypercube hop

$$a \longmapsto a\oplus e_i, \qquad i\in\{1,2,3\}.$$

This is the standard e-cube rule of hypercube networks: compare source and target, flip one differing coordinate at a time, and finish in at most three local hops [6]. In the holonet that same XOR flip is also a register operation, because the chart address is a finite "which-register" word on the photon's self-entangled carrier. The local routing primitive is therefore simultaneously

$$\text{bit flip} \quad = \quad \text{qutrit-frame update} \quad = \quad \text{optical path choice.}$$

A packet in the photonic holonet is not merely a payload. The natural packet header is

$$\mathcal{P} = (\text{payload, Pauli frame, chart word, apartment hop, mirror slot, clock phase}).$$

The payload is the quantum state or teleported gate; the Pauli frame is the two-trit feedforward record; the chart word is the $\mathbb{F}_2^3$ route inside the current cube; the apartment hop selects the next chart; the mirror slot chooses the D_{12} transport bus of §5; and the clock phase is the cyclic C_{12} interrupt that keeps the selector coherent. This is why the same object can be called a computer architecture and a network architecture. A node never sends data to a processor: the act of transport is the processor's gate action.

Theorem 4.2 (The fabric is the building)**.** *The chart-adjacency web (*540 *nodes, two charts adjacent when one's axis is a disjoint transversal pair of the other) is* 6*-regular with exactly* 1620 *edges, and these edges are in canonical bijection with the* 1620 *apartments of the Tits building of* $\mathrm{Sp}(4, \mathbb{F}_3)$*. The web has diameter* 5*; its adjacency module contains the Steinberg representation twice and presses against the Ramanujan bound only on a* 15*-dimensional sentinel eigenspace.* Witnesses: `bt777`, `bt779`, `bt744`.

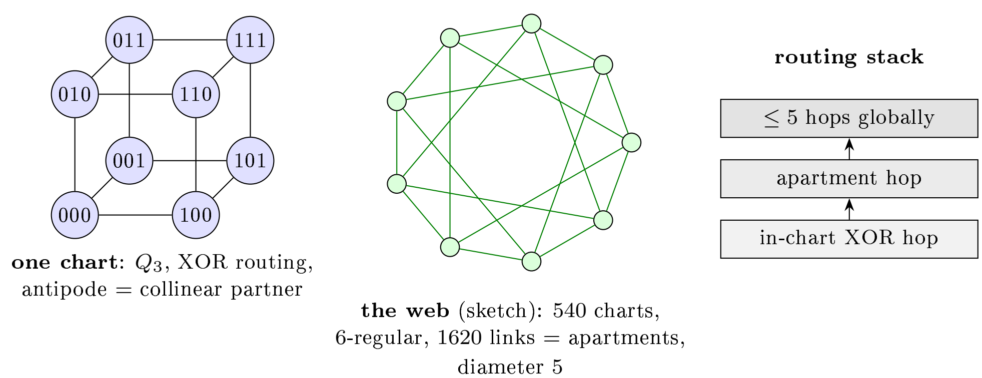


Figure 3: Network schematics. Left: a single chart with its $\mathbb{F}_2^3$ addressing (XOR routing native). Center: the 540-chart web whose links are the apartments of the Tits building (sketched as a circulant fragment). Right: the two-level routing stack.

Routing a packet means choosing XOR moves inside charts and apartment hops between them; since chart moves *are* register operations (§6), routing is computation. At scale the recursion is the holonet ladder: 40^n cores at level n, the same $\mathrm{SRG}(40, 12, 2, 4)$ at every level, with the level-0 chart's $1 + 12 + 27$ degree split realized as self pole, gauge shell, and matter shell of each node.

Corollary 4.3 (Two-plane routing bound)**.** *Inside a chart the worst route length is* 3*, the diameter of* Q_3*. Between charts the worst route length is* 5*, the diameter of the apartment web. One recursive digit of a holonet address therefore costs at most*

$$3 + 5 = 8$$

reversible transport moves. BT827 globalizes this to $8n = 8 \log_{40} N$ *for* $N = 40^n$ *photonic leaves.*

5 The middleware: tomotope mirrors and the runtime bus

Definition 5.1 (Tomotope packet)**.** *In this paper the tomotope is used in the only way the finite calculation currently justifies: as a local middle-layer incidence packet. A tomotope packet has four vertex carriers, twelve local edge axes, sixteen triangular face labels, and forty-eight edge–face incidence blocks. The doubled base/shadow fiber gives* 192 *flags. Thus the tomotope is the durable packet format sitting between the fast cube router and the global mirror bus; it is not introduced as an ambient continuous torus, and it is not identified with the rectangle clock.*

The atlas contains two different 2160-element boundary worlds. They must not be collapsed. The rectangle side is cyclic: its stabilizer is C_{12} and it supplies the selector's phase clock. The chart-transversal side is dihedral: its stabilizer is D_{12} and it supplies the mirror transport bus. Equal cardinality is the red herring; different stabilizer structure is the theorem.

Theorem 5.2 (The D_{12} mirror bus)**.** *The* 540 *cube charts carry exactly four common-transversal slots each, so the global chart-transversal space has* $540 \cdot 4 = 2160$ *slots. The map*

$$(\textit{chart},\ \textit{common transversal line}) \longmapsto (\textit{chart},\ \textit{antipode pair cut out by that transversal})$$

is an equivariant bijection onto the atlas antipode-slot space. Its projective stabilizer has order 12 *and GAP identifies it as* D_{12}*, not* C_{12}*.* Witness: `bt815_global_2160_transversal_gset.py`.

Locally, a skew-line chart has four common isotropic transversals. Read those four transversals as the four vertex carriers of the tomotope middle layer. Each transversal is a K_4, and each K_4 has three opposite-edge axes and four triangular faces. Thus the chart carries

$$4 \text{ vertices}, \qquad 4 \cdot 3 = 12 \text{ edge labels}, \qquad 4 \cdot 4 = 16 \text{ face labels},$$

and the middle incidence packet has

$$4 \cdot 3 \cdot 4 = 48$$

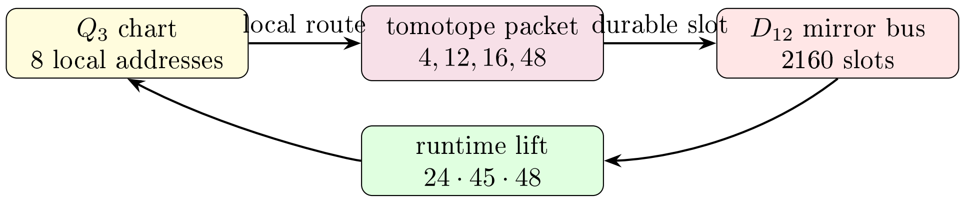
edge–face blocks, with the doubled base/shadow fiber restoring 192 flags. This is the local $\langle r_0, r_3 \rangle$ tomotope block layer, not an analogy to a toroidal polyhedron. *Witness:* *`bt814_tomotope_middle_layer_from_residual_tetrahedra.py`*.

Theorem 5.3 (Photonic mirror middleware)**.** *The full optical Clifford runtime factors as*

$$|\mathrm{Sp}(4, \mathbb{F}_3)| = 51840 = 24 \cdot 2160 = 24 \cdot 45 \cdot 48.$$

Here $24 = f$ *is the full* $\mathrm{Sp}(4,3)$ *lift of the projective* D_{12} *mirror-slot stabilizer,* 45 *is the hyperbolic polar-pair/tritangent geography, and* 48 *is the local tomotope edge–face middle layer (also the order of the chart group* O_h*). Equivalently, the photon is clocked by the cyclic selector side and routed by the dihedral mirror side; the complete Clifford group is the lift of that mirror bus through the tomotope middle carrier.* Witness: `bt826_photonic_mirror_middleware.py`.

This is the architectural closure missing from a bare universality claim. The tritter, phase plate, and controlled delay generate the whole Clifford group, but the group is *organized* as a middleware stack: full slot stabilizer, global mirror bus, polar vacuum geography, and tomotope local block. In software terms, C_{12} is the clock interrupt, D_{12} is the mirror router, 48 is the local packet format, and $24 \cdot 45 \cdot 48$ is the finite runtime.



fast XOR routing is committed through the tomotope middle layer and reflected by the mirror bus.

Figure 4: Middleware schematic. The cube is the reversible local router; the tomotope is the edge–face packet format; the D_{12} bus is the global mirror transport object.

6 The software: braids, teleported gates, universality

6.1 Exact braid words

In the anyonic Fibonacci representation $\sigma_1 = \mathrm{diag}(\zeta^4, -\zeta^2)$ ($\zeta = e^{i\pi/5}$) one has *exactly* $\sigma_1^5 = Z$ and $\sigma_1^{10} = I$ — not merely projectively (verified in the cyclotomic ring $\mathbb{Z}[\zeta_{10}]$). In the dual ($\pm$) encoding of the local $K_{3,3}$ cycle code $\mathbb{F}_2^4$ this makes every register bit-flip an exact five-letter braid word, and the flat global register (the unique connected gluing with trivial holonomy across the atlas) carries the action coherently: a zero-Berry-phase memory. *Witnesses:* `bt740`*,* `bt741`*.*

6.2 The photon is its own gate

A gate U is stored as the self-entangled state $(I \otimes U)|\Omega\rangle$; Bell-projecting an input register against the photon's *past* register outputs U (times a known Pauli correction) on the *future* register — verified for all nine outcomes. The machine's gate library is a library of self-entangled states; executing software means interfering the photon with its own history. *Witness:* `bt821`*.*

6.3 Instruction set architecture

The practical instruction set is finite:

$$\mathcal{I}_{\mathrm{holo}} = \{F_p, F_f,\ S_p, S_f,\ \mathrm{CX}_{p\to f},\ \sigma^5 = Z,\ D_{12}\text{-mirror},\ M_{36}\text{-magic}\}.$$

F and S generate single-qutrit Clifford frame changes; CX couples past to future; $\sigma^5 = Z$ gives exact braid-register flips; the D_{12} mirror operation moves a packet through the atlas without confusing it with the cyclic selector clock; and M_{36} denotes injection from the thirty-six magic rays of §9. The Clifford part normalizes Pauli frames and is efficiently classically trackable; it is not universal by itself. Universality starts only when the Clifford runtime can consume the intrinsic magic sector. This is the engineering meaning of "matter equals magic": the same thirty-six rays that appear as the matter shell are the non-Clifford fuel for the processor.

6.4 The minimal classical shadow

The smallest classical headline compatible with the substrate is the pair

$$(\lambda, q) = (2, 3).$$

This is the same two-state, three-symbol signature made famous by the Wolfram–Smith weakly universal Turing machine. The present paper does not use that result as the proof of quantum universality; the proof is the Clifford closure plus magic/contextuality [8, 7]. The point is sharper: the classical shadow of the photonic holonet lands on the same minimality boundary. The quantum machine has forty rays and an $81 = q^4$ protected register, but its smallest visible finite-state control alphabet is already the substrate pair $2, 3$.

6.5 The Universality Theorem

Theorem 6.1 (Clifford completeness from optics)**.** *The symplectic images of the photon's physical elements — the tritter F_3 on either register, the quadratic phase plate S, and the delay-conditioned* CX *— generate all of* $\mathrm{Sp}(4, \mathbb{F}_3)$ *(closure of order* 51840*, exact). Hence tabletop optics realize the complete two-qutrit Clifford group modulo Paulis and phases.* Witness: `bt825_universality_theorem.py`.

Theorem 6.2 (Universality)**.** *Clifford completeness, together with the machine's intrinsic magic supply (§9) and the strict positivity of its contextual fraction* ($1/10$*, exact), implies universal quantum computation by magic-state injection; for qutrits, contextuality is necessary and sufficient for magic distillation (Howard–Wallman–Veitch–Emerson) [3]. One self-entangled photon on the* $W(3, 3)$ *mesh is a universal quantum computer whose transport is its gate action.*

7 Fractal scaling: the computer is the network

The single-core theorem is not the end of the architecture. The substrate is self-similar: any node may be replaced by a fresh copy of the same forty-node holonet, with the parent line/point incidence becoming the routing grammar for the child layer.

Definition 7.1 (Recursive holonet)**.** *Let $\mathcal{H}_0$ be one self-entangled photonic qutrit. Define $\mathcal{H}_n$ recursively by replacing each point of one copy of* $W(3, 3)$ *with a copy of $\mathcal{H}_{n-1}$. Thus $\mathcal{H}_1$ is the single-core machine of this paper and $\mathcal{H}_n$ is a* 40*-ary fractal computer/network.*

Theorem 7.2 (BT827 holonet scaling law)**.** *A level-n holonet has*

$$N_n = 40^n$$

photonic leaf cores and

$$I_n = 1 + 40 + \cdots + 40^{n-1} = \frac{40^n - 1}{39}$$

total $W(3, 3)$ *instances. Its total edge-qutrit slots are* $240I_n$*, its directed transport slots are* $480I_n$*, its chart routers are* $540I_n$*, its apartment links are* $1620I_n$*, its mirror slots are* $2160I_n$*, and its local Clifford runtime atoms are* $51840I_n$*. Worst-case reversible routing across the recursive address uses at most*

$$8n = 8 \log_{40} N_n$$

moves, where $8 = 3 + 5$ *is one in-chart hypercube diameter plus one chart-web apartment diameter. Durable commits use the distinct Császár/tomotope clock*

$$T(0) = 1, \qquad T(g) = 4(7^g - 1) \quad (g \geq 1).$$

Witness: `bt827_holonet_fractal_architecture.py`.

level n	leaves 40^n	$W(3,3)$ instances	mirror slots	runtime atoms	route bound
1	40	1	2160	51840	8
2	1600	41	88560	2125440	16
3	64000	1641	3544560	85069440	24
4	2560000	65641	141784560	3402829440	32

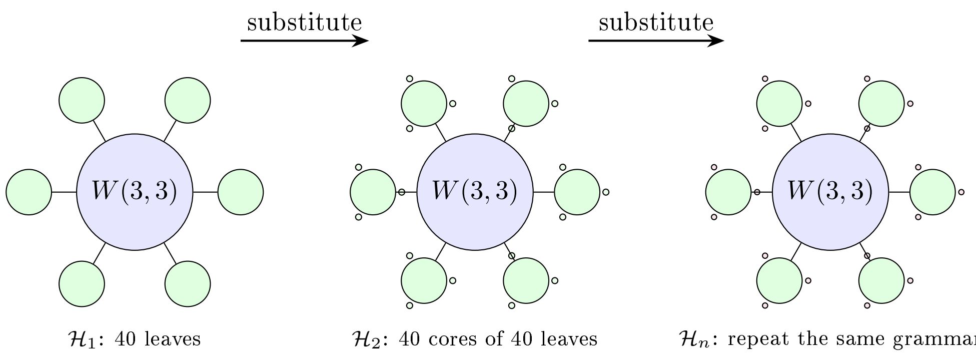


Figure 5: Fractal scaling schematic. The drawn fans show only a few children; the rule is exact substitution of forty child holonets at each substrate point. Routing grows logarithmically in the number of photonic leaves because each recursive digit costs at most eight reversible moves.

This resolves the computer-engineering question. A conventional machine grows by adding processors connected by a separate network. The holonet grows by adding copies of the same object; each copy contains its own processor, router, memory, clock, and packet format. There is no von Neumann bus to saturate. At every scale, the edge set is both communication bandwidth and entangling-gate availability, the chart atlas is both routing table and instruction decoder, and the Steinberg 81-sector is both protected memory and fault-tolerant logical space.

8 Runtime closure: compile, monitor, commit

The architecture now has an executable operating-system loop.

$$\text{address word} \longrightarrow \text{packet route} \longrightarrow \text{sentinel monitor} \longrightarrow \text{durable tomotope commit.}$$

BT828–BT831 make each arrow finite and testable.

Theorem 8.1 (BT828 packet compiler)**.** *A recursive address word with digits in* $\{0,\ldots,39\}$ *compiles to a sequence of digit packets*

$$(Q_3\text{-}XOR\ hops,\ apartment\ hops,\ D_{12}\ mirror\ slot,\ C_{12}\ phase,\ tomotope\ block).$$

Each digit uses at most three XOR hops and at most five apartment hops, hence at most eight reversible moves. The stress route

$$(0,7,14,21,28,35) \longrightarrow (39,32,25,18,11,4)$$

compiles to 47 *reversible moves, below the level-six bound* 48. Witness: `bt828_holonet_packet_compiler.py`.

Theorem 8.2 (BT829 sentinel monitor)**.** *The projector onto the $g = 15$ adjacency sector of $W(3,3)$ has exact entry profile*

$$P_{-4}(x,x) = \frac{3}{8}, \qquad P_{-4}(x,y) = \begin{cases} -\frac{1}{8}, & x \sim y, \\ \frac{1}{24}, & x \nsim y. \end{cases}$$

Consequently a full context line K_4 is sentinel-invisible, while localized, mirror, and shell faults activate the sentinel with exact energies:

fault	*energy*	*normalized fraction*
single point	$3/8$	$5/13$
adjacent edge	$1/2$	$5/19$
nonedge mirror pair	$5/6$	$25/57$
gauge shell $N(x)$	6	$5/7$
matter shell	$27/8$	$5/13$

The immune system is therefore not generic noise detection: it is blind to legal context-line traffic and tuned to off-context routing and shell congestion. Witness: `bt829_fault_sentinel_monitor.py`.

Theorem 8.3 (BT830 two-phase commit clock)**.** *The runtime has two clocks. The reversible prepare phase finishes in at most $8n$ moves. The durable commit phase waits for*

$$T(n) = 4(7^n - 1).$$

For $1 \leq n \leq 24$, $T(n)$ is always divisible by 24 and by 8, so it is aligned with both the local runtime stabilizer and the single-digit route window. But it is not always divisible by the full $8n$ route epoch: the first desynchronization is $n = 5$, with remainder $24 = f$. Thus the commit layer is deliberately slower and cover-like; it is not merely the route clock renamed. Witness: `bt830_two_phase_commit_clock.py`.

8.1 The infinite-cover warning

The tomotope contributes a harder boundary than the earlier packet picture showed. Monson, Pellicer, and Williams prove that the tomotope has infinitely many distinct minimal regular covers; for coprime odd $p, q > 1$ there are minimal covers R_p, R_q of the tomotope such that neither covers the other, and neither covers the special R_2 cover [11]. They also record the toroidal cover arithmetic

$$|\mathrm{Mon}(Q_k)| = 36864\, k^6, \qquad Q_k : \ (4, 24, 32, 8, 4)k^3$$

for vertices, edges, triangles, tetrahedra, and octahedra.

Architecturally this is a big correction. The 48-block tomotope middle layer is a stable local ABI, but the global regular cover is not unique. Durable storage must therefore carry a *cover index* k as an implementation gauge. Fast routes compile to the same invariant packet, while persistent storage may choose one minimal cover family without invalidating the other non-comparable families. *Witness: `bt831_tomotope_minimal_cover_architecture.py`.*

8.2 Cover-indexed storage, rerouting, and guard bands

The infinite-cover warning would be useless as engineering unless it changed the runtime. BT832–BT834 make it operational. The principle is that the local tomotope packet is the ABI and the chosen regular cover is the storage backend.

Theorem 8.4 (BT832 cover-indexed durable storage). *For every supported cover index k, durable storage is the lifted packet space*

$$\{0,\dots,47\}\times\mathbb{Z}_k^3, \qquad |slots| = 48k^3,$$

with projection back to the fast-route ABI by forgetting the $\mathbb{Z}_k^3$ coordinate. The kernel to the base tomotope packet has order

$$k^6 = (k^3)^2,$$

matching the Monson–Pellicer–Williams toroidal cover law $|\mathrm{Mon}(Q_k)| = 36864k^6$. Changing k changes durable capacity and kernel size, but not the packet program seen by the compiler. Tested cover gauges $k = 3, 5, 7, 11, 13$ all reduce exactly to the same 48-block ABI, and larger k reduces the load fraction of the same stress program. Witness: `bt832_cover_indexed_durable_storage.py`.

Theorem 8.5 (BT833 sentinel-aware packet rerouting). *The BT829 projector supplies a compiler-visible cost function. Before the durable commit, each digit route may insert up to two W33 waypoint points and minimize*

$$\textit{sentinel energy first, reversible move count second.}$$

The extra reversible moves are charged to the slower BT830 commit phase, not to the BT827 fast $8n$ route bound. In the level-six stress program the direct route uses 47 moves and total sentinel energy 4; the sentinel-aware route uses 58 moves, still far below the 470592-tick commit window, and lowers the total sentinel energy to 2. Each tested digit route strictly lowers its own $g = 15$ energy; adjacent endpoints can be context-completed to zero, and nonedge mirror pairs drop below their direct $5/6$ activation. Witness: `bt833_sentinel_aware_packet_rerouter.py`.

Theorem 8.6 (BT834 desynchronization guard-band arithmetic). *The two-phase clock has an exact number-theoretic boundary. Full route-epoch synchronization is*

$$8n \mid T(n) = 4(7^n - 1),$$

equivalently

$$2n \mid 7^n - 1.$$

For every odd prime power $p^a \mid n$ with $p \neq 7$, this requires $\mathrm{ord}_{p^a}(7) \mid n$; if $7 \mid n$, synchronization is impossible because the base is not coprime. The first failure is $n = 5$: $\mathrm{ord}_5(7) = 4 \nmid 5$, and the remainder is

$$T(5) \bmod 40 = 24 = f.$$

Thus the first cover-index where durable storage and the full route epoch separate is not empirical drift; it is the $f = 24$ guard band of the tomotope-backed runtime. Witness: `bt834_desync_guard_band_arithmetic.py`.

Theorem 8.7 (BT838 tomotope Wythoff runtime ladder). *The Grünbaum–Coxeter/tomotope operation tables give the local tomotope counts*

$$\textit{original, rectified, truncated, maximal expanded, omnitruncated} \quad = \quad 4,\ 12,\ 24,\ 48,\ 96.$$

These are exactly the runtime packet-growth layers:

$$4 = \mu \quad (\textit{vertex carriers}), \qquad 12 = k \quad (\textit{edge axes}), \qquad 24 = f \quad (D_{12}\ \textit{lift}),$$

$$48 = 2f \quad (\textit{BT814 packet ABI}), \qquad 96 = 4f \quad (\textit{half-flag boundary}).$$

The doubled omnitruncated boundary gives the verified full packet flags, $2 \cdot 96 = 192$*. After a cover lift this becomes*

$$48k^3 = \textit{BT832 lifted packet capacity}, \qquad 192k^3 = |W_k|,$$

the W_k *order already recorded in the tomotope cover architecture. Thus the Wythoff operations are not decorative variants; they are the compiler's packet expansion states:*

$$\textit{carrier} \to \textit{edge-axis} \to \textit{mirror lift} \to \textit{packet ABI} \to \textit{flag boundary}.$$

Witness: `bt838_tomotope_wythoff_runtime_ladder.py`.

9 The fuel: matter = magic

Theorem 9.1 (The triality)**.** *In the photon's two-qubit reading, exactly the* 4 *basis rays are stabilizer states and all* 36 ω*-rays are magic (no stabilizer state has support* 3*). Consequently three classifications coincide exactly, on rays* $[4, 36]$ *and on contexts* $\{4{:}1,\ 1{:}12,\ 0{:}27\}$*: the holonet vacuum split, the Schmidt (self-entanglement) strata, and the magic strata.* The matter shell is the machine's non-classical resource sector. Witness: `bt822`.

Theorem 9.2 (Three grades and the cube inside)**.** *Stabilizer fidelity splits the* 36 *magic rays into grades*

$$\underbrace{8}_{2^q} \ \textit{deep}\ \left(F = \tfrac{2+\sqrt{3}}{6}\right), \qquad \underbrace{24}_{f} \ \textit{mid}\ \left(F = \tfrac{5+2\sqrt{3}}{12}\right), \qquad \underbrace{4}_{\mu} \ \textit{shallow}\ \left(F = \tfrac{3}{4} = \tfrac{q}{\mu}\right),$$

the shallow grade sitting exactly at the Werner decoherence threshold. The 8 *deep rays are precisely the point set of a hyperbolic polar pair (the* 24*-cell vacuum axis), and the* 27 *fully-magic contexts stratify by grade signature as* $1+8+12+6$ *— the cell census of the cube.* Witnesses: `bt822`, `bt823`.

Theorem 9.3 (Exact contextuality budget)**.** *The maximum number of contexts a classical* $\{0, 1\}$ *valuation can satisfy exactly-once is exactly* $36 = (q!)^2$ *(complete branch-and-bound over all* 2^{40} *markings): classicality saturates at the spread count, the deficit is exactly* $\mu = 4$*, and the contextual fraction is exactly* $1/10 = 1/\Phi_4$*. A perfect valuation would be an ovoid, and* $W(3, 3)$ *has none: the true independence number is* $\alpha = 7 = \Phi_6$*, attained only by the beacon heptads of §12, against the Lovász bound* $\vartheta = 10$ *— the* $\vartheta - \alpha$ *gap* $3 = q$ *is the combinatorial face of Kochen–Specker.* Witnesses: `bt818`, `bt823`.

10 Memory, protection, and the immune system

Theorem 10.1 (Steinberg memory)**.** *The Levi-graph cycle space and the chart-overlap* 81*-sector are both irreducible and equal to the Steinberg representation of* $\mathrm{PSp}(4, 3)$ *— the Solomon–Tits homology of the building, whose canonical basis is the* 81 *apartments through any chamber. Schur's lemma then forces the uniqueness of the chart–cycle bridge. Restricted to a Sylow* 3*-subgroup the module is free of rank one: the protected memory is one free generator over the substrate's ternary symmetry.* Witnesses: `bt742`, `bt744`.

The error layer is native: the $[[240, 81, 4, 3]]_3$ CSS code on the edge qutrits has all four parameters substrate-primitive; the dual Császár/Szilassi cross-check supplies parity-free error detection; and the web's unique non-Ramanujan eigenspace — dimension $15 = g$ — is the spectral sentinel where drift appears first.

11 Time: three clocks and a quasicrystal

Theorem 11.1 (Clock trichotomy). *Of the program's three natural clocks, exactly one is internal:* $\mathbb{Z}_{12}$ *(the rectangle clock,* $12 \mid 51840$*) runs the selector and duo machinery, while* $\mathbb{Z}_7$ *(Császár) and* $\mathbb{Z}_{13}$ *(Singer) do not divide the group order and act as external references — both with decimal period* $6 = q!$*, so the digit expansion of* $1/7$ *is literally the internal clock's shadow modulo its duo bit.* Witnesses: `bt774`, `bt803`, `bt807`, `bt819`.

Theorem 11.2 (The Boerdijk–Coxeter time quasicrystal). *Driving the loop by the BC twist* $\theta = \arccos(-2/3)$ *per round trip yields a stroboscopic orbit that provably never repeats (Niven), with the Steinhaus three-distance signature at all substrate step counts and — the substrate's signature moment — exactly* two *gap lengths at* $n = 30 = h(E_8)$*, the BC ring length, where closure happens in* S^3 *(the* 600*-cell) rather than the phase circle (deficit* 0.0158*). This is the photonic sibling of the Fibonacci-drive dynamical topological phase.* Witness: `bt820`.

Theorem 11.3 (The arrow of time). *Without its two-trit feedforward record, the photon's self-configuration loop is the completely depolarizing channel with unique fixed point* I/q *— exactly the Bell marginal; with the record it is unitary. Decoherence is self-forgetting, priced at* $\log_3 q^2 = 2$ *trits per cycle.* Witness: `bt823`.

12 Synchronization: beacons and schedules

The maximum classically coherent constellations are the *beacon heptads*: seven states with all pairwise overlaps exactly $1/3 = 1/q$ — a complete K_7 interference mesh, the Császár toroidal node realized in light. There are exactly 2880 heptads in a single transitive family; each has a $\mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry fixing a unique *master beacon* and cycling two triads ($7 = 1 + 3 + 3$), and single-beacon swaps connect the family into its own exchange network: the reference frames of the network form a network. The measurement timetable is rigid: there exist exactly 36 complete schedules (partitions of the 40 states into 10 contexts) — an exhaustive result that simultaneously proves every spread of $W(3,3)$ is regular — with each context appearing in exactly 9 schedules. *Witnesses: `bt817`, `bt819`.*

Theorem 12.1 (Schedule-overlap law). *Two distinct schedules share exactly* 1 *or exactly* $4 = \mu$ *contexts: each schedule has* 15 near *partners (overlap* 4*) and* 20 far *partners (overlap* 1*), the two relation classes of the double-six diagonal* $[1, 15, 20]$*, with the exact counting identity* $1 \cdot 20 + 4 \cdot 15 = 80 = 10 \times (9 - 1)$*. Operationally: a cheap timetable switch preserves* 4 *of* 10 *calibrated contexts and a far switch only* 1*, so a controller hopping among near partners re-calibrates* 6 *contexts per hop — the timetable library is itself a metric space whose geometry is the double-six relation.* Witness: `bt835_schedule_overlap_theorem.py`.

The Grünbaum–Coxeter cells. Each schedule also has an internal shape. The full schedule stabilizer S_6 is 2-homogeneous on the 10 contexts, but its icosahedral core A_5 — the 600-cell level of the platonic ladder, and *exactly* the automorphism group of the hemi-dodecahedron $\{5,3\}_5$ — splits the $\binom{10}{2} = 45$ context pairs as $45 = 15 + 30$, and the 15-orbit graph on the 10 contexts is the **Petersen graph**: the edge skeleton of the hemi-dodecahedron, the cell of Coxeter's exceptional 57-cell. Dually, the hidden 6-set of the S_6 action realizes the hemi-icosahedron $\{3,5\}_5$ — the cell of Grünbaum's 11-cell — with incidence count $(6, 15, 10)$: six hidden points, fifteen duads, ten triple-splits = the contexts themselves. The exceptional polytopes' groups do not embed in $\mathrm{Sp}(4, \mathbb{F}_3)$, but they are substrate arithmetic ($|\mathrm{PSL}(2,11)| = 660 = k \cdot N_{\text{eff}}$, $|\mathrm{PSL}(2,19)| = 3420 = k \cdot g \cdot 19$, Heawood genus $H(19) = 20$ = the BC ring count, and $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$): the full 11-cell and 57-cell live elsewhere, yet their *cells* are carried by every measurement timetable of this machine, glued by the substrate's own group instead of $\mathrm{PSL}(2,p)$. *Witness:* `bt836_gc_hemicells_in_spreads.py`.

Theorem 12.2 (The library is a classical geometry)**.** *The timetable library carries its own rank-3 geometry, and the hemicell structure fibers over it with perfect uniformity: (i) every skew line pair lies in exactly* 3 *schedules, so the* (schedule $\supset$ pair) *flag count is* $36 \times 45 = 1620$ *— the apartment count of the Tits building — and* 540×3*; (ii) the near-partner graph on the* 36 *schedules is strongly regular* $\mathrm{SRG}(36, 15, 6, 6)$*, the* other *classical rank-3 graph of* $U_4(2) \cong \mathrm{PSp}(4, \mathbb{F}_3)$*, so the control plane (timetables) and the data plane (*$W(3,3) = \mathrm{SRG}(40, 12, 2, 4)$*) are the two rank-3 geometries of one group; (iii) each schedule stabilizer contains exactly* 6 *icosahedral* A_5 *cores (*$216 = 6^3$ *in all, each marking its unique schedule via its* $[10, 30]$ *line orbits) giving* 6 *distinct Petersen splits; each internal pair is a Petersen edge under exactly* 2 *of the* 6 *cores, hence every skew pair of* $W(3,3)$ *has exactly* $6 = 3 \times 2$ *hemi-dodecahedral homes (*$3240 = 540 \times 6$ *Petersen flags).* Witness: `bt837_schedule_library_geometry.py`.

Theorem 12.3 (Two compasses, twelve Petersens, $4 \cdot K_{10}$)**.** $\mathrm{PSp}(4, \mathbb{F}_3)$ *has exactly two conjugacy classes of* A_5*, both with normalizer* S_5 *and* 216 *conjugates, unfused by the outer automorphism, and* both fiber over the same 36 schedules*: every schedule stabilizer contains* $12 = 6 + 6$ *icosahedral cores —* 6 duad *cores (line signature* $[10, 30]$*, the Theorem-12.2(iii) cores) and* 6 pentad *cores (signature* $[5, 5, 10, 20]$*: transitive on the schedule, plus two distinguished* pentads *of* 5 *pairwise-disjoint lines outside it). The two* 216*-sets are non-isomorphic G-sets (rank* 10 *suborbits* $[1, 5, 10, 10, 20, 20, 20, 30, 40, 60]$ *vs* $[1, 5, 10, 10, 20, 20, 30, 30, 30, 60]$*); the Petersen flags form the coset geometry* $\mathrm{PSp}(4, \mathbb{F}_3)/D_4$ *(*3240 *flags, dihedral stabilizer). There are* 432 *distinct pentads in* two chiral orbits *of* 216 *(stabilizer order* 120*); every core pairs one LEFT with one RIGHT pentad, and each pentad serves exactly one core. Pentads are* maximal partial spreads*: no schedule contains one. The two pentads of one core interlock as* $K_{5,5}$ *minus a perfect matching — mutual disjointness is* forbidden *by the overlap law (it would create a* 0*-overlap schedule) — and the deleted matching consists of* 5 skew *pairs, i.e. five hypercube charts; over the* 216 *cores these matchings cover the* 540*-chart atlas **exactly twice** (*$1080 = 540 \times 2$*): the routing fabric is readable off the pentad compasses, five charts per needle. Finally,* all twelve *cores split the* 45 *internal pairs as Petersen* 15*-orbits, and the twelve Petersen edge-sets cover every pair exactly* $\mu = 4$ *times:*

$$4 \cdot K_{10} \;=\; 12 \ \textit{Petersens}.$$

Schwenk's classical obstruction (no edge-partition of K_{10} *into* 3 *Petersen graphs) dissolves at the substrate multiplicity* μ*. The numerical coincidence* $3240 = \#\{\textit{triangles of } Q\}$ *is refuted at the G-set level (orbits* $[360, 2880]$ *vs transitive).* Witnesses: `bt843_icosahedral_compass.py`, `bt844_double_five_and_six_` `bt845_pentad_exchange_and_chart_double_cover.py`.

Theorem 12.4 (The amalgamation ladder)**.** *The* 11*-cell and* 57*-cell are* universal *abstract polytopes: the unique ones with hemi-icosahedral facets and hemi-dodecahedral vertex figures (group* $L_2(11)$*, no proper quotients) and vice versa (*$L_2(19)$*). Their incidence is icosahedral-subgroup geometry: each of* $L_2(11)$*,* $L_2(19)$*,* $U_4(2)$ *has exactly two conjugacy classes of* A_5*, and in all three the distinguished incidence is* intersection in D_{10} *with valency* 6 *— the* 11*-cell's vertex–facet relation, the* 57*-cell's facet adjacency (the Perkel graph), and the substrate's compass incidence (*$216 + 216$*, bipartite* 6*-regular,* $216 \times 6 = 1296 = 6^4$ *incident pairs =* 36 *schedules* $\times$ 36 *core pairs). The closures of one* *amalgam* $A_5 *_{D_{10}} A_5$ *(GAP):*

$$\langle A, B\rangle = \begin{cases} L_2(9) \cong A_6\ (360) & \textit{in } U_4(2)\textit{: inside exactly one schedule stabilizer,} \\ L_2(11)\ (660) & \textit{the } 11\textit{-cell,} \\ L_2(19)\ (3420) & \textit{the } 57\textit{-cell.} \end{cases}$$

One local icosahedral amalgam, a PSL(2)*-ladder of completions over* $9 = q^2$*,* $11 = k - 1$ *(the Ihara prime),* 19 *(Heawood,* $H(19) = 20$*). Universality plus Lagrange (*$11, 19 \nmid 25920$*) shows the substrate carries the* complete local data *of both exceptional polytopes while their rank-*4 *amalgamation is arithmetically forbidden inside* $\mathrm{Sp}(4, \mathbb{F}_3)$*: the GC polytopes are the substrate's sibling completions, not subobjects. One step up the universal ladder (*$\{\{5,3\},\{3,5\}_5\}$*,* 10006920 *facets) the completion group is* $J_1 \times L_2(19)$ *— the first sporadic Janko group, whose involution centralizer* $2 \times A_5$ is *the compass datum. Finally, the compass Petersen* 15*-orbit carries exactly* 12 *pentagons splitting under the compass into two* 6*-orbits, each covering every edge exactly twice: two* chiral fully-faced *hemi-dodecahedra per needle, not merely skeletons.* Witnesses: `bt848_universal_amalgam_ladder.py`, `.tmp/gap_bt848*.g`; cf. Hartley's classification (17 universal rank-4 locally projective polytopes, 441 quotients) and Hartley–Leemans on $\{5, 3, 5\}$.

The barren rung and the maniplex workaround. Relaxing polytopes to *maniplexes* (flag graphs without the intersection property; the tomotope middleware of this machine is a uniform 4-polytope whose monodromy $18432 = 96 \times 192$ fails that property, giving it *infinitely many* distinct minimal regular covers — the canonical rank-4 cover pathology) does not lift the obstruction: exhaustive search shows $U_4(2)$ admits *no* rank-4 regular maniplex with a 5 in its type, and the amalgam closure $A_6 = L_2(9)$ admits no regular maniplex at any rank — it is the *barren rung* of the PSL(2) ladder, whose populated rungs are exactly the 11-cell (q=11), the 57-cell (q=19), and Leemans–Toledo's degenerate single-facet $\{5, 5, 5\}$ family (our control search rediscovers all of them, including both exceptional polytopes). This is why the Grünbaum–Coxeter content of the substrate is *delocalized* over the 36 schedules — Petersen splits, chiral pentads, chart covers, dark dodecahedra — rather than crystallized in one flag object, and why the machine's runtime uses the tomotope: it is the flag structure that survives the obstruction. The tomotope's symmetry type is now exact: $|\mathrm{Aut}| = 96$ (independent centralizer computation on the 192 flags), two flag orbits $[96, 96]$, **class** $2_{\{0,1,2\}}$ — only the cell-color crosses orbits, and the orbits are precisely "flag in a tetrahedron" vs "flag in a hemioctahedron": the middleware *alternates spherical and projective cells*, the exact situation the classical "locally X" taxonomy cannot name, and the setting of Mochán's 2-orbit polytopality theory. *Witnesses:* *`bt849_maniplex_barren_rung.py`*, *`bt850_tomotope_two_orbit_class.py`*, *`.tmp/gap_bt849*.g`*.

Reconstruction, the chart K_5, and the dark dodecahedra. The pentad core's full signature $[5, 5, 10, 20]$ (lines), $[20, 20]$ (points) is rigid: both chiral pentads cover the *same* 20-point orbit (lit/dark halves of the substrate), the five matching charts' common-transversal bundles overlap to exactly the 10 schedule lines — **a pentad pair generates its timetable** — and the line $\to$ chart-pair map is a bijection onto the $\binom{5}{2} = 10$ edges of K_5: the schedule *is* a K_5 on its own charts (4 transversals per chart, 2 charts per line). The 20 dark lines complete the $\{5, 3\}$ family: their 190 pairs split $[10, 30, 30, 30, 30, 60]$ under the core, and exactly two of the 30-orbits are **dodecahedron skeletons** (3-regular, girth 5, diameter 5) — a chiral pair of genuine dodecahedra beneath every compass needle, one level under the hemi-dodecahedral Petersen splits of the schedule itself. *Witnesses:* *`bt846_pentad_core_anatomy.py`*, *`bt847_dark_dodecahedron.py`*.

Theorem 12.5 (BT839 GC operation Euler/flag audit)**.** *Treating the full Grünbaum–Coxeter note as a data table gives a second invariant. The base tables for the* 11*-cell,* 57*-cell, tomotope partial a, and tomotope partial b are all Euler-neutral:*

$$V - E + F - C = 0.$$

But every Wythoff operation table has a fixed family charge:

$$\chi_{\mathrm{op}}(11\text{-}cell) = -11, \qquad \chi_{\mathrm{op}}(57\text{-}cell) = -57, \qquad \chi_{\mathrm{op}}(tomotope) = -4.$$

The omnitruncated rows are full-flag carriers:

$$660 = |\mathrm{PSL}(2,11)| = 11 \cdot A_5 = kN_{\mathrm{eff}},$$

$$3420 = |\mathrm{PSL}(2,19)| = 57 \cdot A_5 = k \cdot g \cdot 19,$$

and the tomotope has 96 *half-flags, whose double is the verified* 192*-flag BT814 packet. The* 57*-cell also locks to the W33 timetable library: BT837 gives* 3240 *Petersen homes, and*

$$3240 + k \cdot g = 3240 + 180 = 3420.$$

Thus one $k \cdot g$ sentinel sheet completes the W33 Petersen-home carrier to the 57*-cell flag count.*

The regular-polychoron alignment supported by the cell/symmetry data is

$$11\text{-}cell \to 600\text{-}cell, \qquad tomotope \to 24\text{-}cell, \qquad 57\text{-}cell \to 120\text{-}cell.$$

The $57 \to 120$ *link is direct at the dodecahedral cell level; the* $11 \to 600$ *link is weaker but real, through hemi-icosahedral local structure and the quotient* $2I \to A_5$ *rather than through a direct tetrahedral-cell match. The swapped* 11/57 *orientation is recorded as a lower-scoring dual/vertex-figure hook, not discarded.* Witness: `bt839_gc_operation_euler_flag_audit.py`.

Theorem 12.6 (BT840–BT842 carrier closure)**.** *The three unresolved carriers in the Grünbaum–Coxeter paragraph are now explicit objects.*

(i) The* 180 *sentinel sheet. *The verified Clifford L/R boundary has* 36 *cells arranged as a* 6×6 *rook grid. Its zero-overlap relation is not a numerical afterthought: it is the union of the* 6 *row fibres and* 6 *column fibres, each fibre carrying the duads of a six-set. Hence*

$$12 \cdot \binom{6}{2} = 12 \cdot 15 = k \cdot g = 180.$$

This is exactly the sheet missing from BT837:

$$3240 + 180 = 3420 = |\mathrm{PSL}(2,19)|.$$

Equivalently, the timetable library gives 216 *Petersen cores* ($216 \cdot 15 = 3240$)*, and the rook boundary supplies the missing* 12 *six-set fibres* ($12 \cdot 15 = 180$)*.*

(ii) The* 660 *carrier. *At an apex cell* (r, c) *of the same* 6×6 *grid, remove the incident row and column. The apex plus the five nonincident rows and five nonincident columns gives*

$$11 = 1 + 5 + 5.$$

Crossing these eleven labels with the verified 60*-element Clifford* A_5 *selector gives*

$$11 \cdot A_5 = 11 \cdot 60 = 660 = kN_{\mathrm{eff}}.$$

This is a carrier theorem only: it does not assert an internal $\mathrm{PSL}(2,11)$ *action on* $W(3,3)$.

(iii) The* 96 *tomotope half-flags. *In the* D_4*-root model of the* 24*-cell, every edge lies in a unique central hexagon plane. The two endpoint axes determine the third, missing axis in that plane, so each edge maps to a Reye incidence* (*missing axis, hexagon plane*)*. Each of the* 48 *Reye incidences has exactly two edge lifts:*

$$96 = 2 \cdot 48.$$

Thus the tomotope omnitruncated half-flag count is literally the 24*-cell edge lift of the common tomotope/Reye spine.*

The hexagon clue supplies the conceptual bridge to the 11*-cell. Each central* 24*-cell hexagon is a six-root cycle* C_6*. Completing that cycle to a full* K_6 *gives* 15 *duads, exactly the hemi-icosahedral skeleton count of the* 11*-cell cell. Across the* 16 *central hexagons,*

$$16\binom{6}{2} = 16 \cdot 15 = 240,$$

the W33 edge count and the E_8 *root count. The tested dot profile of these* 240 *duad slots is*

$$96_{\langle x,y\rangle=1} \;+\; 96_{\langle x,y\rangle=-1} \;+\; 48_{\langle x,y\rangle=-2},$$

namely live 24*-cell edges, mirror pairs, and repeated opposite-axis Reye incidences.* Witnesses: `bt840_clifford_rook_sentinel_sheet.py`, `bt841_eleven_cell_a5_boundary_carrier.py`, `bt842_tomotope_24`

13 Build sheet and falsifiable witnesses

Components (all standard quantum optics): one heralded single-photon source; one polarizing beam splitter; one symmetric 3-port coupler (tritter); a $0/\tau/2\tau$ delay ladder; one bin-synchronous electro-optic modulator; one polarization rotator set to $\arccos(-2/3)$ in the recirculation loop; single-photon detectors.

Witnesses (kill criteria, all parameter-free):

witness	predicted value	substrate form
trace–Choi visibility of F_3	$1/3$	$1/q$
trace–Choi visibility of X, Z	0	—
Kochen–Specker classical budget	$36/40$	$(q!)^2/v$
beacon-mesh pair visibility	$1/3$ (all 21 pairs)	$1/q$
Werner separability threshold	$3/4$	q/μ
BC-drive gap census at n=30	2 gap lengths	$h(E_8)$ ring
D_{12} mirror-slot stabilizer	order 12	projective bus
runtime factorization	$24 \cdot 45 \cdot 48$	$\lvert\mathrm{Sp}(4,\mathbb{F}_3)\rvert$
one recursive route digit	≤ 8 reversible hops	$3+5$
level-n leaves	40^n	v^n
level-n $W(3,3)$ instances	$(40^n-1)/39$	geometric tower
durable commit clock	$4(7^g-1)$ ticks	tomotope/Csaszar
compiled stress route	$47 < 48$ moves	BT828 level six
cover-lifted packet capacity	$48k^3$ slots	BT832 storage gauge
sentinel-aware stress energy	$4 \to 2$	BT833 reroute
full route sync criterion	$2n \mid 7^n - 1$	BT834 guard band
tomotope Wythoff runtime ladder	$4, 12, 24, 48, 96$	BT838 packet growth
GC operation Euler charges	$-11, -57, -4$	BT839 table audit
sentinel projector trace	15	g sector
first full-route desync	$n = 5$ remainder 24	f guard band
tomotope cover ABI	48 blocks / 192 flags	cover-invariant

A measured deviation in any row falsifies the corresponding layer.

14 Implications, corrections, and the ethos

Physics. The vacuum is five-fold degenerate, indexed by the Schläfli inventory; the choice $1 + 12 + 27$ is a parabolic gauge choice, not a necessity. The generation triples of the matter shell are the tritangent triangles of the 27-line configuration, reconstructed here purely from double cosets. Matter and magic coincide: the non-classical resource of the machine *is* its matter sector, with the cube's cell census $1 + 8 + 12 + 6$ grading the fuel.

Computing. Universality requires no exotic hardware: the Clifford layer is three optical elements, the magic layer is the photon's own matter shell, and the topological layer rides the exact $\sigma^5 = Z$ braid identity. The network-is-computer thesis is not an analogy but the statement that the atlas's transport groupoid and the register's gate groupoid are the same groupoid. The middleware theorem adds the missing operating-system layer: the cyclic C_{12} clock orders phase, the dihedral D_{12} mirror bus routes state, the tomotope 48-packet supplies local edge–face incidence, and the full runtime is exactly $24 \cdot 45 \cdot 48 = 51840$.

Networking. The holonet is a universal network because it has addresses, routers, routes, packets, clocks, and error signals at the same level as it has gates. The Q_3 charts give local e-cube routing; the Tits-building apartments give inter-chart routing; the D_{12} mirror slots give transport identity; and the tomotope gives a durable packet shape. Recursive substitution gives 40^n photonic leaves with $O(\log N)$ reversible diameter. The network scales because every child core inherits the same finite protocol rather than depending on a separate global controller.

Physics-to-architecture dictionary. The point-parabolic split $1 + 12 + 27$ is the processor's local I/O anatomy: self pole, gauge neighbors, matter/magic non-neighbors. The edge code on 240 qutrits is bandwidth and memory at once. The $81 = q^4$ Steinberg sector is the protected logical register. The thirty-six spread schedules are the measurement operating system. The five vacua are boot modes. The Boerdijk–Coxeter irrational loop is the watchdog clock that refuses to fall into a finite phase-cycle trap. These are engineering claims only because they are first incidence claims.

The ethos. Two pre-existing corpus claims failed under exact computation and were corrected at their sources during this program: the independence number of $W(3, 3)$ is 7, not 10 (no ovoid exists, the "perfect graph" block is withdrawn), and the Kochen–Specker optimum is 36/40, not 34/40. In both cases the corrected values are *more* substrate-natural than the claims they replaced ($\alpha = \Phi_6$; budget $= (q!)^2/v$) — the pattern a real structure shows when probed honestly.

A Verification ledger

Every theorem above is backed by an executable script in `analysis/` with results in `data/`; the chain BT739–BT839 was developed and pushed incrementally with exact arithmetic (GF(p) elimination, cyclotomic integers, complete branch-and-bound, GAP witnesses for group-theoretic facts).

layer	primary witnesses
duality / carriers	bt817, bt820, bt821
atlas / building / web	bt773, bt777, bt744, bt779
mirror middleware / tomotope runtime	bt814, bt815, bt826
braids / registers	bt740, bt741
vacua / geography / transitions	bt808–bt813, bt816
fuel / contextuality	bt818, bt822, bt823
memory / Steinberg	bt742, bt744
clocks / quasicrystal / arrow	bt774, bt819, bt820, bt823
sync / schedules / overlap law	bt817, bt819, bt835
Grünbaum–Coxeter hemicells / library geometry	bt836, bt837, bt839
two compasses / $4 \cdot K_{10} = 12$ Petersens / chart double cover	bt843–bt845
pentad anatomy / schedule reconstruction / dark dodecahedra	bt846, bt847
amalgamation ladder / universality / J_1	bt848
maniplex barren rung / tomotope workaround / class $2_{\{0,1,2\}}$	bt849, bt850
universality	bt825
fractal computer/network	bt827
runtime compiler / sentinel / commit / cover boundary	bt828–bt834, bt838
corpus corrections	bt818, bt823, bt824

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